

Foundations of Deep Learning

Lecture 7

GENERALIZATION I

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Fall 2025

Problem of interest

Hypothesis space Set $\mathcal{F}$ of functions from $\mathcal{X} \subseteq \mathbb{R}^d$ to $\mathcal{Y} \subseteq \mathbb{R}$

Risk Given a **data distribution** $\mathcal{D}$ over $(\mathcal{X}, \mathcal{Y})$, we define the true risk as

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}}[\ell(f(\mathbf{x}), y)]$$

We will mostly discuss a classification setting where

$$\ell((\mathbf{x}, y), f) = \mathbf{1}_{f(\mathbf{x}) \neq y}.$$

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Empirical risk Typically do not have access to the exact data distribution but only a **sample set** $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n \sim \mathcal{D}$
 $\leadsto$ define the empirical risk as

$$R_S(f) = \sum_{(x, y) \in S} [\ell(f(\mathbf{x}), y)].$$

What's a generalization bound?

The typical form of a generalization bound we are interested in is:

$$\underbrace{R(f)}_{\text{True risk}} \leq \underbrace{R_S(f)}_{\text{Empirical risk}} + \underbrace{m(\text{complexity of class of functions}, n)}_{\text{(a function that approaches 0 as } n \text{ approaches infinity)}},$$

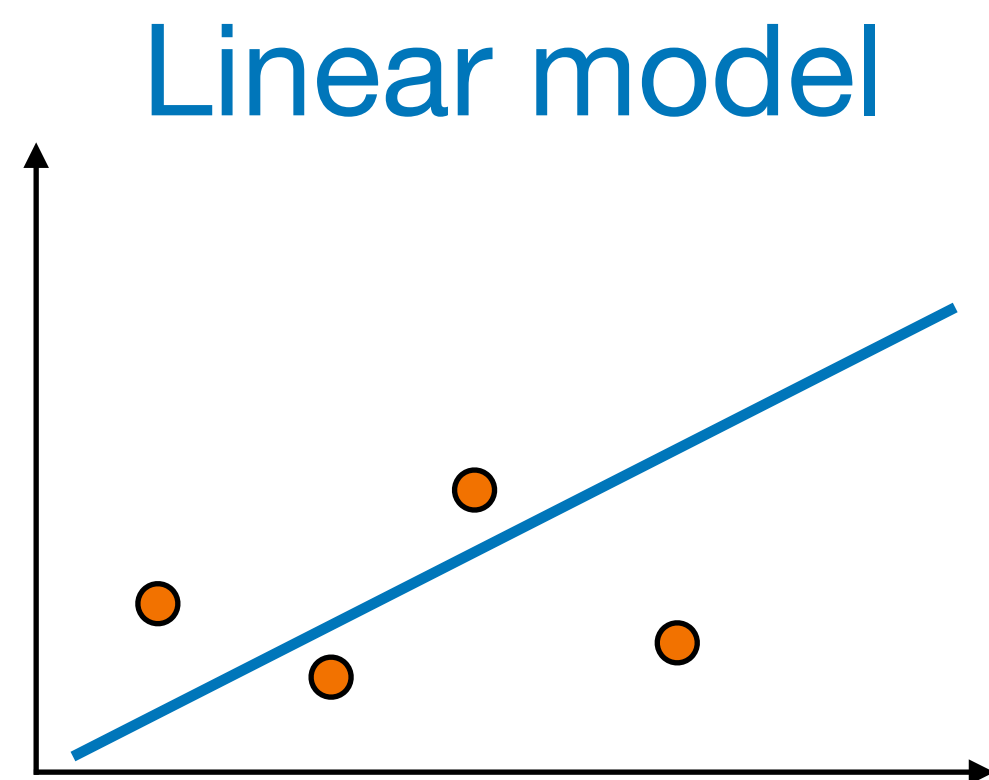
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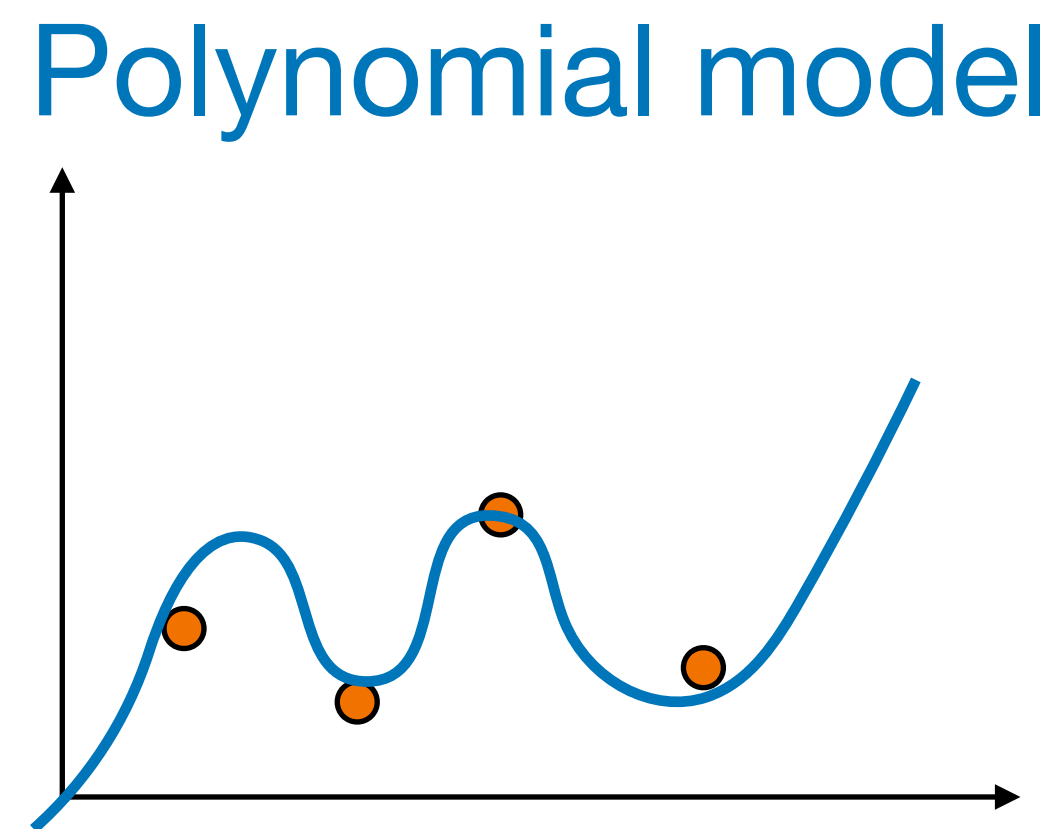
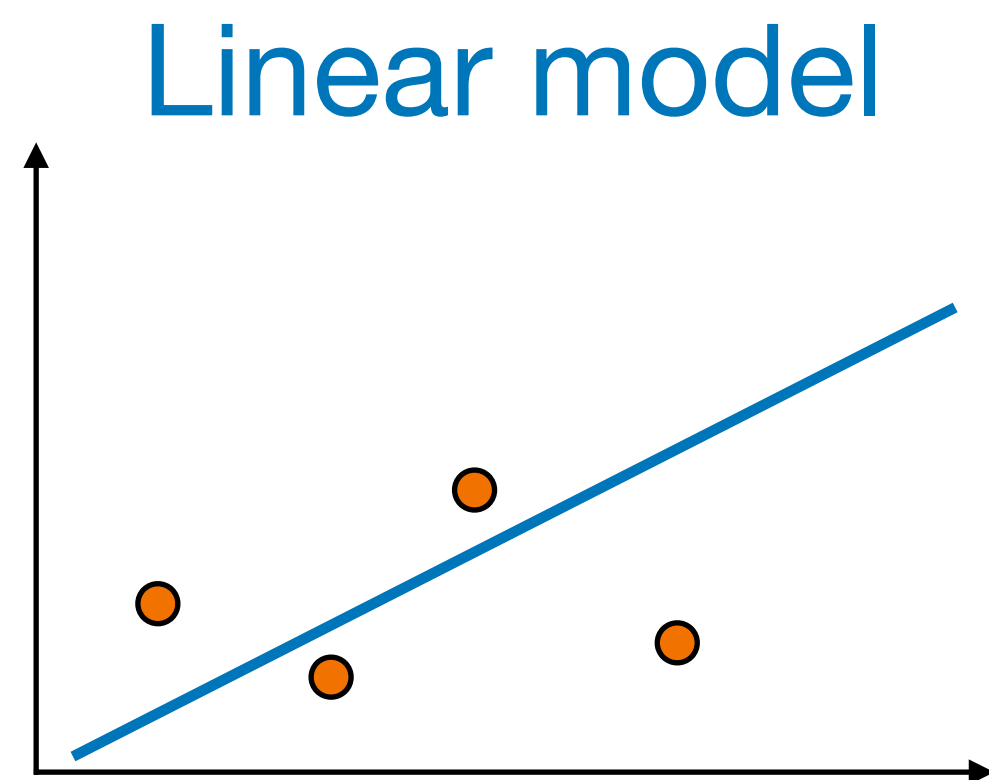


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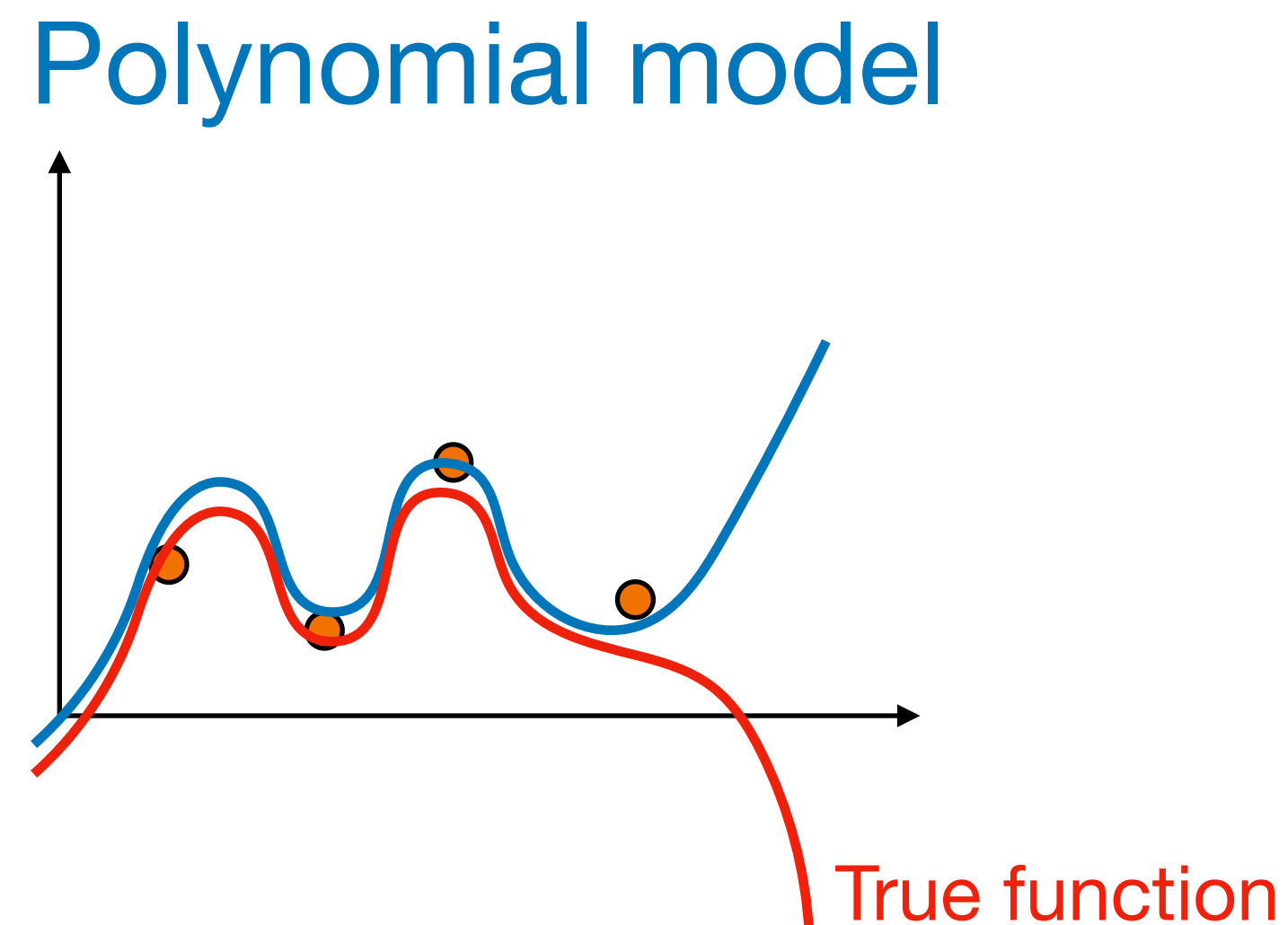
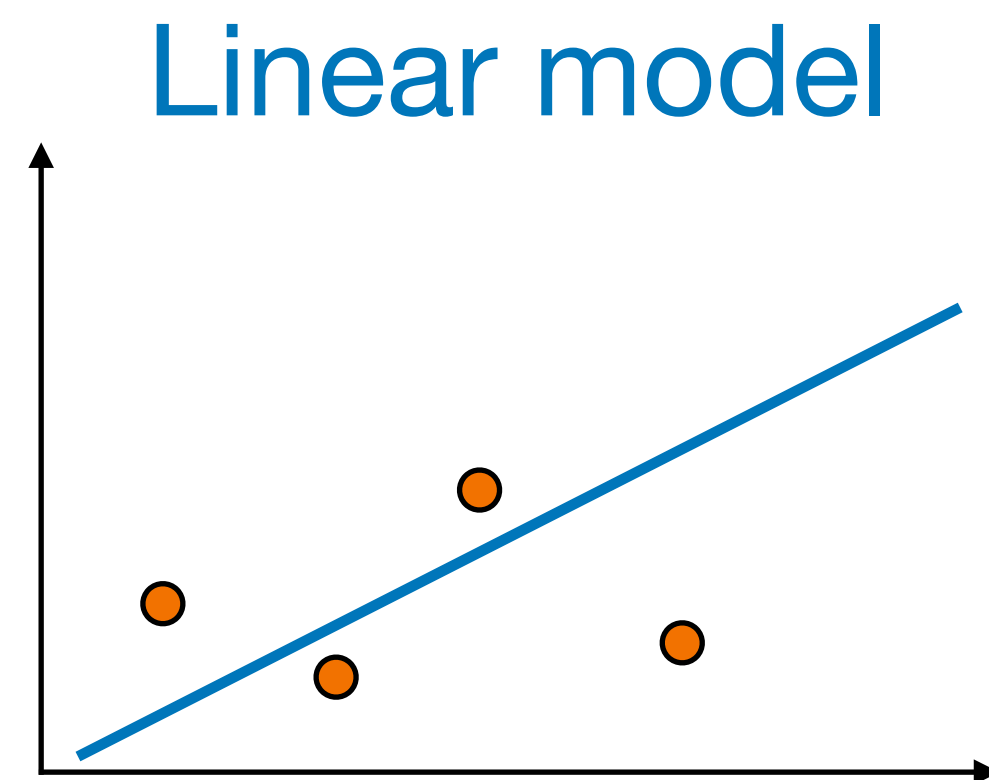


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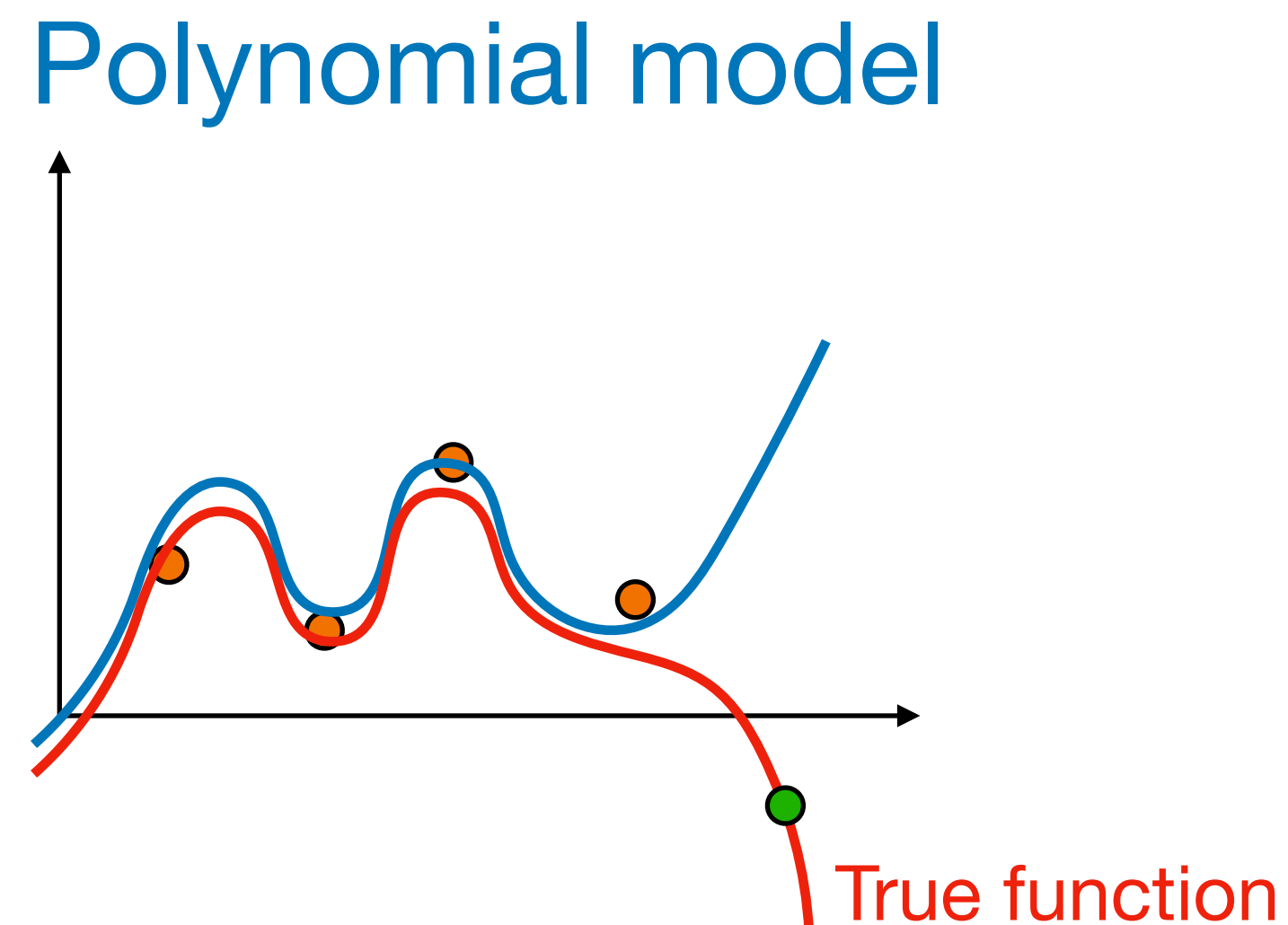
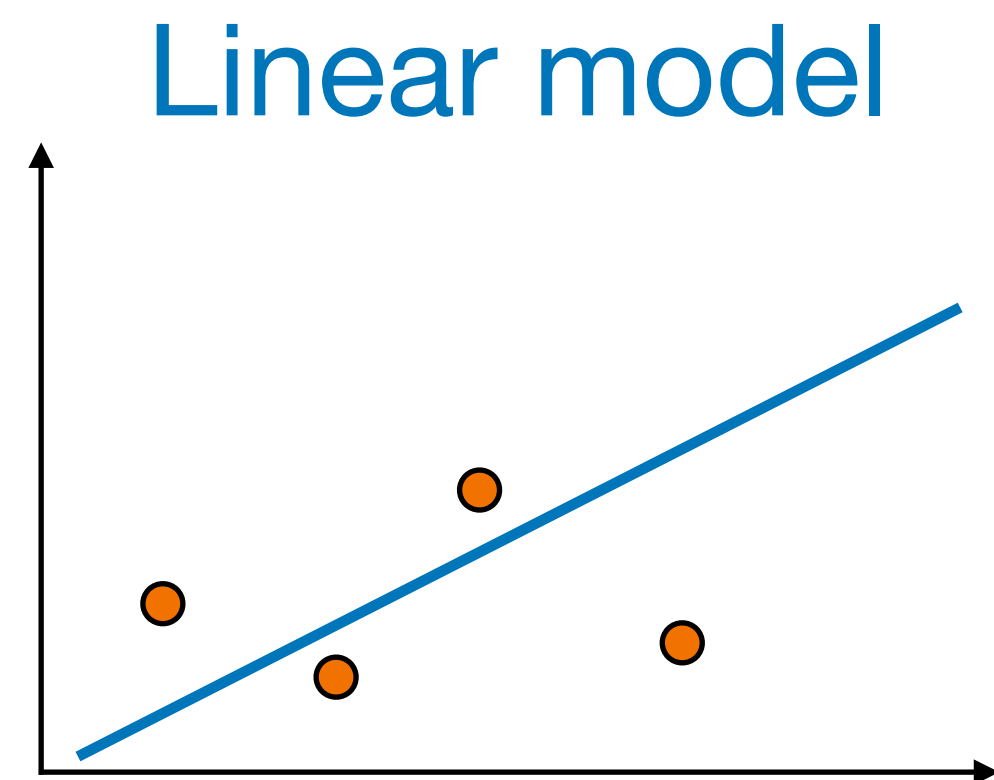


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Section 1

BASIC CONCENTRATION BOUND

Applying Hoeffding's inequality

Let $Z_i = (\mathbf{x}_i, y_i)$ be i.i.d. random variables with $\ell(Z) \in [a, b]$. The quantity we are interested in is

$$R(f) - R_S(f) = \mathbb{E}[\ell(Z)] - \frac{1}{n} \sum_{i=1}^n \ell(Z_i).$$

Applying Hoeffding's inequality

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Using Hoeffding's inequality, as well as the independence of the random variables $\ell(Z_i)$, we get:

$$P \left[\left| \mathbb{E}[\ell(Z)] - \frac{1}{n} \sum_{i=1}^n \ell(Z_i) \right| > \epsilon \right] \leq 2 \exp \left(-\frac{2n\epsilon^2}{(b-a)^2} \right).$$

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Setting the right hand side to be δ , we obtain:

$$\delta = 2 \exp \left(-\frac{2n\epsilon^2}{(b-a)^2} \right) \implies \epsilon = (b-a) \sqrt{\frac{\log \frac{2}{\delta}}{2n}}$$

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Union bound

Therefore, with probability at least $1 - \delta$, for a given $f \in \mathcal{F}$,

$$|R_S(f) - R(f)| \leq (b - a) \sqrt{\frac{\log \frac{2}{\delta}}{2n}}.$$

→ Result is for **one** given element $f \in \mathcal{F}$.

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Generalization to all functions in $\mathcal{F}$

Use union bound:

$$P(C_1 \cup \cdots \cup C_N) \leq \sum_{n=1}^N P(C_n) \leq N\delta.$$

Union bound

We get with probability at least $1 - \delta$,

$$\forall f \in \mathcal{F}, R(f) \leq R_S(f) + (b - a) \sqrt{\frac{\log N + \log \frac{2}{\delta}}{2n}}.$$

Proof...

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Proof...

Uncountable case

What if the set $\mathcal{F}$ is uncountable?

→ Previous approach does not directly work.

There are several **measures of capacity** or size of classes of function:

- ▶ Distribution independent: e.g. VC dimension and growth function
- ▶ Distribution dependent: VC entropy

Section 2

UNIFORM BOUNDS

Principle behind Uniform bounds

Problem:

- ▶ Previous results: for each (fixed) function $f \in \mathcal{F}$, there is a collection of sample sets S for which the bound holds
- ▶ However these sets S may be different for different functions f
- ▶ In other words, for the observed sample, only some of the functions in $\mathcal{F}$ will satisfy the bound.
- ▶ This means that f cannot change with different draws of S for the bound to hold.

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Solution: consider uniform bounds, i.e. bounds that hold uniformly over all functions in the class:

$$\sup_{f \in \mathcal{F}} R(f) - R_S(f).$$

Generalization bound

Theorem 1

Given a set S of size n and $\mathcal{F} = \{f_1, \dots, f_N\}$, let $f^* = \operatorname{argmin}_{f \in \mathcal{F}} R(f)$ and $f_S = \operatorname{argmin}_{f \in \mathcal{F}} R_S(f)$. Then

$$R(f_S) \leq R(f^*) + 2\sqrt{\frac{\log N + \log \frac{2}{\delta}}{2n}}$$

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Proof

Section 3

INFINITE CASE: VC BOUND

VC theory

Can we extend our result to the case where the class $\mathcal{F}$ is infinite?

VC Vapnik & Chervonenkis argued that what really matters for a sample $S = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ is the set

$$\mathcal{F}_{\mathbf{z}_1, \dots, \mathbf{z}_n} = \{(f(\mathbf{z}_1), \dots, f(\mathbf{z}_n)) \mid f \in \mathcal{F}\}.$$

$\rightsquigarrow$ Size of this set is the total number of possible ways that S can be classified.

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VC theory - Example

Example: binary classification $\mathcal{F}_{\mathbf{z}_1, \dots, \mathbf{z}_n}$ is a set of binary vectors $\subset \{-1, +1\}^n$.

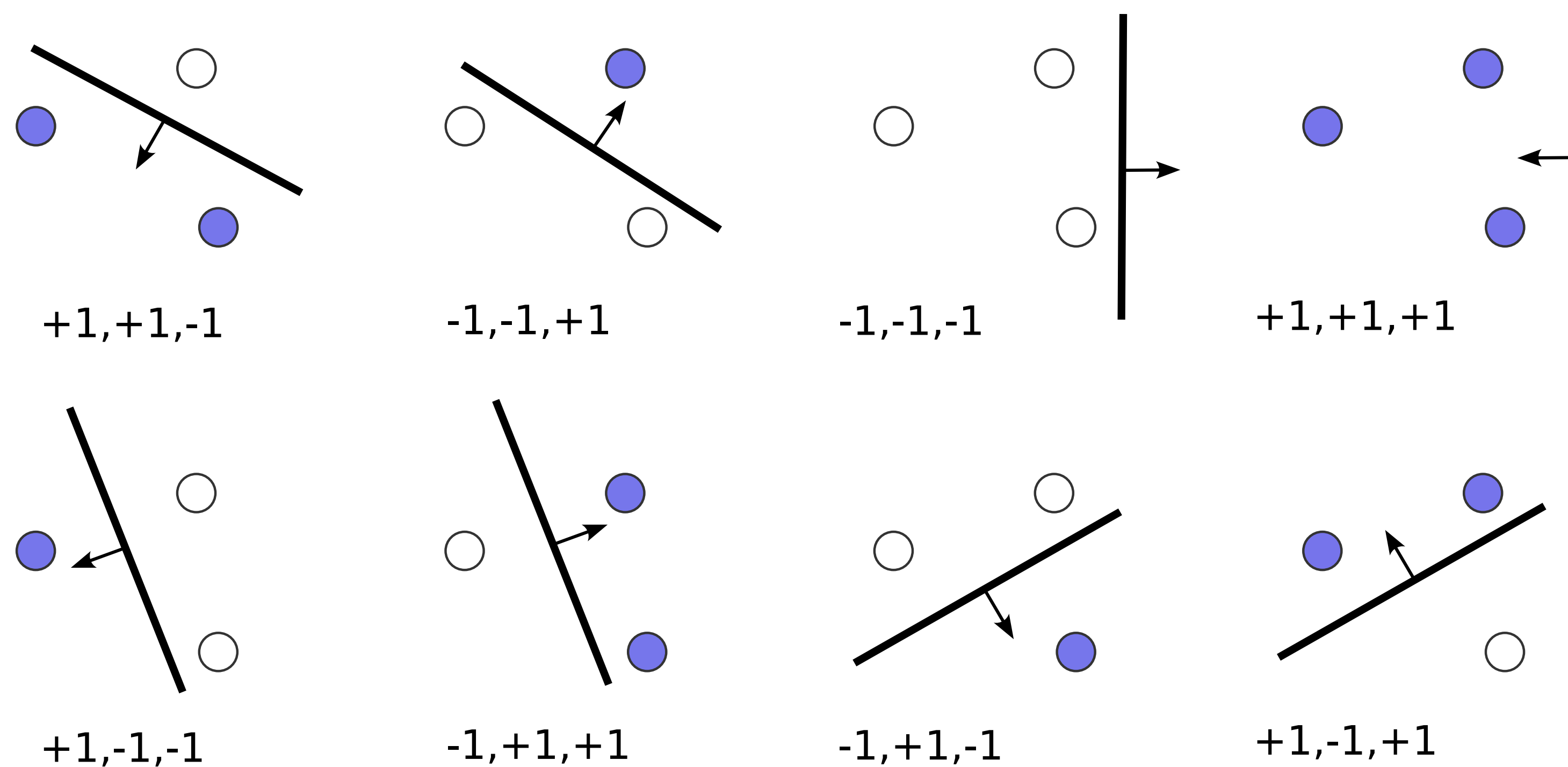


Figure: Illustration of growth function for $n = 3$ datapoints.

VC theory

Definition 2 (Growth Function)

The growth function of $\mathcal{F}$ is equal to

$$S_{\mathcal{F}}(n) = \sup_{(\mathbf{z}_1 \dots \mathbf{z}_n)} |\mathcal{F}_{\mathbf{z}_1 \dots \mathbf{z}_n}|$$

$\rightsquigarrow$ note that we can choose the configuration of the points!

Interpretation Size of this set is the number of possible ways in which the data $\{\mathbf{z}_1 \dots \mathbf{z}_n\}$ can be classified using functions in $\mathcal{F}$. An upper bound on this number is 2^n .

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$$\forall f \in \mathcal{F}, \quad R(f) \leq R_S(f) + 2\sqrt{2 \frac{\log S_{\mathcal{F}}(2n) + \log \frac{2}{\delta}}{n}},$$

where $S_{\mathcal{F}}(2n)$ is the growth function of the function class $\mathcal{F}$ defined as

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Proof.

Proof omitted. See statistical theory book.



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VC dimension - example

Example: VC dimension of linear classifiers in $\mathbb{R}^d$ is $d + 1$.

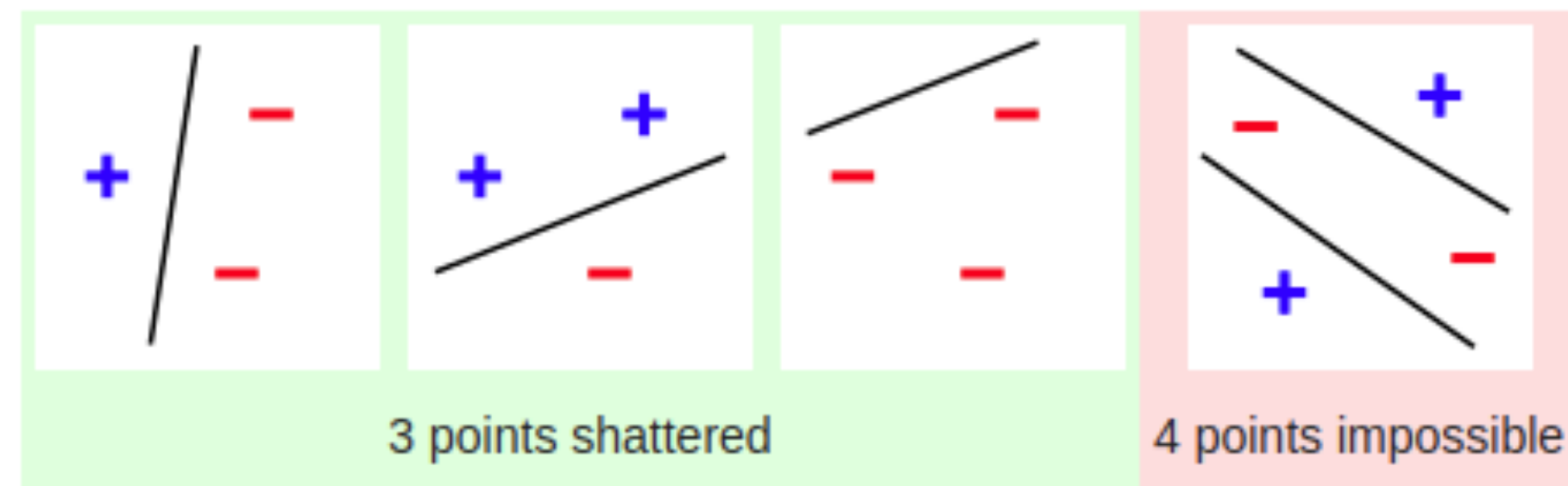


Figure: Illustration of growth function. Source: Wikipedia.

VC dimension

Shattering We say that $\mathcal{F}$ shatters S if $|\mathcal{F}_S| = 2^{|S|}$

VC dimension The VC dimension of a function class $\mathcal{F}$ is the cardinality of the largest set that it can shatter.

Definition 4 (VC dimension)

The VC dimension of a class $\mathcal{F}$ is the largest n such that

$$S_{\mathcal{F}}(n) = 2^n.$$

Sauer-Shelah lemma

Lemma 5 (Sauer-Shelah lemma)

*Let $\mathcal{F}$ be a class of functions with finite VC-dimension h . Then,
 $\forall n \in \mathbb{N}$,*

$$S_{\mathcal{F}}(n) \leq \sum_{i=0}^h \binom{n}{i},$$

and for all $n \geq h$,

$$S_{\mathcal{F}}(n) \leq \left(\frac{en}{h}\right)^h.$$

$\leadsto S_{\mathcal{F}}(n)$ grows very fast as a function of n (faster than exponential), but once the number of datapoints n is larger than the VC dimension h , the growth becomes polynomial.

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Generalization error

One can also use the upper bound on $S_{\mathcal{F}}(n)$ in combination with Theorem 3 in order to obtain the following generalization bound that depends on h and n :

$$\forall f \in \mathcal{F}, \quad R(f) \leq R_S(f) + 2\sqrt{2 \frac{h(\log(2n/h) + 1) + \log \frac{2}{\delta}}{n}}.$$

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VC theory: Application to Neural Networks

Piecewise linear networks: [BHLM19] proves VC dimension is $\mathcal{O}(WL \log(W))$ where W = number of weights and L = number of layers

$\leadsto$ Number of datapoints has to scale inversely proportionally to $\mathcal{O}(WL \log(W))$.

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Remarks

- ▶ Bounds tend to be vacuous for modern deep neural networks that rely on a large number of parameters
- ▶ Partly due to generality of VC bounds

Section 4

RADEMACHER BOUND

Rademacher Complexity

Rademacher random variable: σ_i is defined as

$$\sigma_i = \begin{cases} +1, & \text{with prob. } 1/2 \\ -1, & \text{with prob. } 1/2. \end{cases}$$

Definition 6 (Rademacher complexity)

Let $S = \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$ be a set of samples drawn i.i.d. from D .

Let $\mathcal{F}$ be a class of functions $f : Z \rightarrow \mathbb{R}$. The empirical Rademacher complexity is then defined as

$$\hat{\mathbb{R}}_S(\mathcal{F}) = \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(z_i) \right].$$

Rademacher Complexity

Remarks

- ▶ The supremum measures the maximum correlation between $f(z_i)$ and σ_i over all $f \in \mathcal{F}$
- ▶ Taking the expectation over σ , we can say the empirical Rademacher complexity measures the ability of functions from $\mathcal{F}$ to fit random noise.

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Expected Rademacher complexity Take the expectation of $\hat{\mathbb{R}}_S(\mathcal{F})$ over all samples of size n , i.e.

$$\mathbb{R}_n(\mathcal{F}) = \mathbb{E}_S[\hat{\mathbb{R}}_S(\mathcal{F})],$$

which measures the expected noise fitting ability of $\mathcal{F}$ over all datasets S .

Rademacher-based uniform convergence

Theorem 7 (Rademacher-based uniform convergence)

Fix the distribution D and $\delta \in (0, 1)$. If $\mathcal{F} \in \{f : Z \rightarrow [0, 1]\}$ and $S = \{z_1, \dots, z_n\}$. Then with probability at least $1 - \delta$ over the draw of S ,

$$\mathbb{E}_D[f(z)] \leq \hat{\mathbb{E}}_S[f(z)] + 2\mathbb{R}_n(\mathcal{F}) + \sqrt{\frac{\log(1/\delta)}{2n}}$$

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This is the function not the loss

Proof sketch

Rademacher-complexity: from f to ℓ

Lemma 8

Let $\mathcal{F}$ be a family of functions and let $\mathcal{G}$ be the family of loss functions associated to $\mathcal{F}$ for the zero-one loss:

$\mathcal{G} = \{(x, y) \mapsto \mathbf{1}_{f(x) \neq y} : f \in \mathcal{F}\}$. Let $S_{\mathcal{X}} = (x_1, \dots, x_n)$. Then, the following holds for the empirical Rademacher complexities of $\mathcal{G}$ and $\mathcal{F}$:

$$\mathbb{R}_S(\mathcal{G}) = \frac{1}{2} \mathbb{R}_{S_{\mathcal{X}}}(\mathcal{F}). \quad (1)$$

Rademacher bound for Linear Classes

Lemma 9

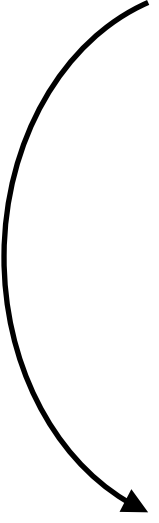
Let $\mathcal{F}$ be the class of linear predictors, with the L_2 -norm of the weights bounded by W_2 . Also assume that with probability one that $\|\mathbf{x}\|_2 \leq D$. Then

$$\mathbb{R}(\mathcal{F}) \leq \frac{DW_2}{\sqrt{n}}$$

Note that in the bound derived in Lemma 9, the Lipschitz constant of f w.r.t. to $\mathbf{x}$ appears, since $\|\nabla_{\mathbf{x}} f(\mathbf{z}_i)\|_2 = \|\mathbf{w}\|_2 \leq W_2$.

$$\mathbb{R}(\mathcal{F}) = \mathbb{E}_{\sigma} \left[\frac{1}{n} \sum_{i=1}^n \sup_{f \in \mathcal{F}} \sigma_i f(z_i) \right]$$

$$\mathcal{F} = \{(\mathbf{w}^\top \mathbf{x}_1, \mathbf{w}^\top \mathbf{x}_2, \dots, \mathbf{w}^\top \mathbf{x}_n) : \|\mathbf{w}\|_2 \leq W_2\}.$$



$$\begin{aligned} \mathbb{R}(\mathcal{F}) &= \mathbb{E}_\sigma \left[\frac{1}{n} \sum_{i=1}^n \sup_{f \in \mathcal{F}} \sigma_i f(z_i) \right] \\ &= \frac{1}{n} \mathbb{E}_\sigma \left[\sup_{\mathbf{w}: \|\mathbf{w}\|_2 \leq W_2} \sum_{i=1}^n \sigma_i \mathbf{w}^\top \mathbf{x}_i \right] \end{aligned}$$

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Linearity

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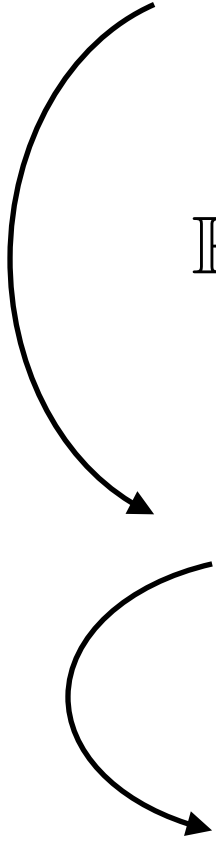
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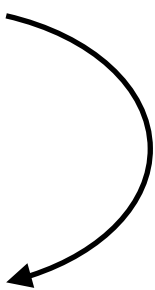
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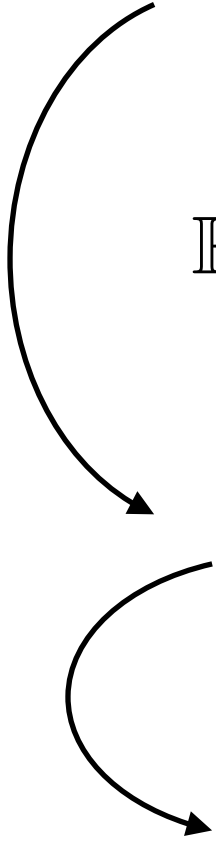
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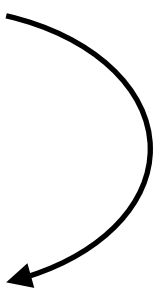
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$\sigma_i = \pm 1$

$\|\mathbf{x}\|_2 \leq D$

Composition Lemma

In neural networks, we often use composition of functions. Can we bound the Rademacher complexity of composition of functions?

Lemma 10 (Composition lemma)

For $\mathcal{F} \subset \mathbb{R}^d$, $\phi : \mathbb{R} \rightarrow \mathbb{R}$, let $\mathcal{F}' := \{\phi \circ f : f \in \mathcal{F}\}$. If ϕ is L -Lipschitz continuous, i.e. $|\phi(t) - \phi(t')| \leq L|t - t'|$, then for any m ,

$$\mathbb{R}(\mathcal{F}') \leq L\mathbb{R}(\mathcal{F}).$$

Proof.

Exercise. □

Application to Neural Networks

One-layer neural network: Based on Lemma 9 and 10, we can derive a bound on the Rademacher complexity of $f(\mathbf{x}) = \sigma(\mathbf{W}\mathbf{x})$ where σ is Lipschitz.

Extension to multiple layers: Use the composition lemma.

Recap

1. Uniform bound: $G.E. \leq 2\sqrt{\frac{\log N + \dots}{2n}}$
2. Growth function: $G.E. \leq 2\sqrt{2\frac{\log S_{\mathcal{F}}(2n) + \dots}{n}}$
3. VC-dimension: $G.E. \leq 2\sqrt{2\frac{h(\log(2n/h) + 1) + \dots}{n}}$
4. Rademacher: $G.E. \leq \mathbb{R}(f)$

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Link to regularization

Application to Neural Networks

Some words of caution:

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This is a very active area of research $\rightsquigarrow$ next time, we will discuss PAC-Bayes bounds.



Peter L Bartlett, Nick Harvey, Christopher Liaw, and Abbas Mehrabian.

Nearly-tight vc-dimension and pseudodimension bounds for piecewise linear neural networks.

The Journal of Machine Learning Research, 20(1):2285–2301, 2019.



Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals.

Understanding deep learning (still) requires rethinking generalization.

Communications of the ACM, 64(3):107–115, 2021.